

CS & IT ENGINEERING



Computer Network

Application Layer

Lecture No. - 05

By - Abhishek Sir





Recap of Previous Lecture



Topic

HTTP

Topic

Email



Topics to be Covered



Topic

Email

Topic

FTP

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





Topic : IMAP

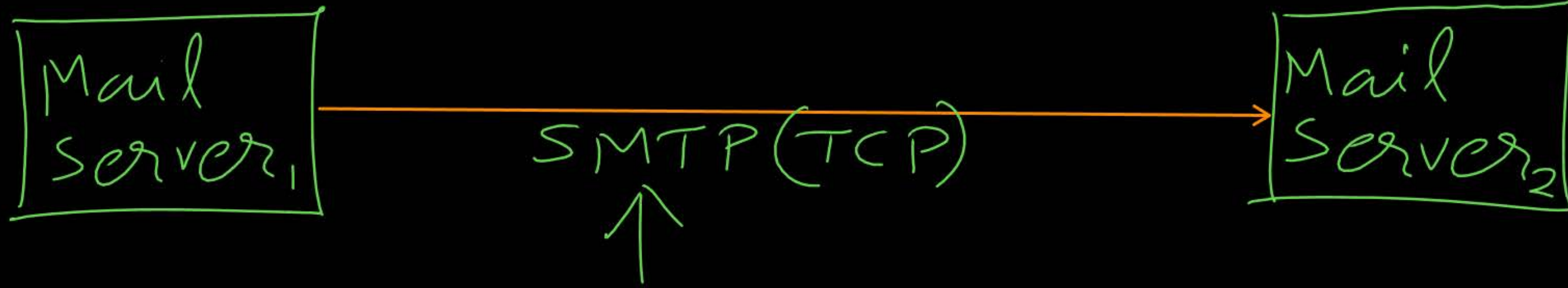


- Application Layer 'Stateful' protocol
- IMAP : Internet Mail Access Protocol
- Retrieving email from mail server
[Message stored on mailbox]
- Earlier Protocol : POP (Post Office Protocol)



*SMTP Uses one TCP connection

*IMAP uses one TCP connection



#Q. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

[GATE 2019, 1-Mark]

IIT-M

~~(A) IMAP, POP3~~

✓ (B) SMTP, POP3

~~(C) SMTP, MIME~~

~~(D) IMAP, SMTP~~

① SMTP

② IMAP (POP)

Ans: B

#Q. Consider different activities related to email:

m1: Send an email from a mail client to a mail server (SMTP)

m2: Download an email from mailbox server to a mail client IMAP/POP

m3: Checking email in a web browser HTTP/HTTPS

Which is the application level protocol used in each activity?

[GATE 2011, 1-Mark]

~~(A) m1: HTTP m2: SMTP m3: POP~~

~~(B) m1: SMTP m2: FTP m3: HTTP~~

(C) m1: SMTP m2: POP m3: HTTP

~~(D) m1: POP m2: SMTP m3: IMAP~~

Ans: C

IIT-M



Topic : FTP



FTP : File Transfer Protocol

→ Reliable transfer of files
[uses TCP as transport protocol]

→ Uses two TCP connection
[one for control and one for data]





Topic : FTP Control Connection

→ FTP client contact FTP server at port 21
[opens persistent TCP control connection]

→ Client obtains authorization over TCP control connection *

→ Client browses remote directory by sending commands over control connection

⇒ Files are not transmit over this connection



Topic : FTP Commands

[FTP Request]



→ ASCII (human-readable format)

→ Sent over control connection

1. USER username

2. PASS password

3. LIST : Return list of files in current directory

4. RETR filename : retrieves (gets) file from server

5. STOR filename : stores (puts) file onto server

⇒ Require
TCP data
connection



Topic : FTP Responses



→ Status Code and Phrase

e.g. 331 username ok, password required



Topic : FTP Data Connection



- FTP client and FTP server opens non-persistent TCP data connection
[for file transfer operation, Port 20 at FTP server]
- In Active mode : FTP server initiate to open TCP data connection
- In Passive mode : FTP client initiate to open TCP data connection



Topic : FTP Data Connection



- After transferring one file, client and server closes TCP data connection
- TCP control connection remains open, for sending commands

#Q. In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

[GATE 2015, Set-1, 1-Mark]

✓
IIT-K

- ✓ (A) HTTP, FTP
- ~~(B) HTTP, TELNET~~
- ~~(C) FTP, SMTP~~
- ~~(D) HTTP, SMTP~~

Ans: A



Topic : FTP



→ Transfer any kind of file

→ Stateful protocol

[FTP server maintains “state” for connected clients]

#Q. Which of the following is/are example(s) of stateful application layer protocols?

(i) HTTP → stateless (ii) FTP → stateful

~~(iii) TCP~~ → stateful transport (iv) POP3 → stateful

[GATE 2016, Set-1, 1-Mark]
IISC

(A) (i) and (ii) only

(B) (ii) and (iii) only

✓ (C) (ii) and (iv) only

(D) (iv) only

Ans: C

#Q. The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are:

TCP

UDP

FTP → TCP

UDP

[GATE 2013, 1-Mark]

IIT-B

- ~~(A) TCP, UDP, UDP and TCP~~
- ~~(B) UDP, TCP, TCP and UDP~~
- ✓ (C) UDP, TCP, UDP and TCP
- ~~(D) TCP, UDP, TCP and UDP~~

Ans: C

#Q. Which one of the following protocols is NOT used to resolve one form of address to another one?

[GATE 2016, Set-1, 1-Mark]

- ~~(A)~~ DNS : Host Name $\longleftrightarrow$ IP Add. 115C
- ~~(B)~~ ARP : IP Address $\longrightarrow$ MAC Add.
- ☒ (C) DHCP $\longrightarrow$ IP Add. allocation dynamic
- ~~(D)~~ RARP : MAC Address $\longrightarrow$ IP Add.

Ans: C



Topic : Stateful and stateless protocol

Transport Layer :-

Stateless : UDP

Stateful : TCP ✓

Application Layer :-

Stateless : DNS, HTTP, SMTP

Stateful : FTP, IMAP (POP), HTTPS, SMTP

SMTP

stateless

stateful



Topic : Syllabus

Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Data link layer : framing, error detection, Medium Access Control, Ethernet bridging; Flow Control

Routing protocols : shortest path, flooding, distance vector and link state routing;

① Introduction

⑤ CRC

⑦

→ CSMA/CD

⑥

④ Routing



Topic : Syllabus



Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT);

② IP Add.
IP Header
③

Transport layer : flow control and congestion control, UDP, TCP, sockets;

⑧ TCP

Application layer protocols : DNS, SMTP, HTTP, FTP, Email.

⑨



2 mins Summary



Topic

Email ✓

Topic

FTP ✓



THANK - YOU

